CS23336/GE23231/GE23233-IPP/PUP/PSPP

Status	Finished
Started	Sunday, 1 November 2026, 10:04 PM
Completed	Sunday, 1 November 2026, 10:13 PM
Duration	8 mins 55 secs
Grade	**9.00** out of 10.00 (**90**%)

Question 1

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer:(penalty regime: 0 %)

```
def binary_search(arr, target):
    arr.sort()

    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return True
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    return False


arr = list(map(int, input().split(',')))
target = int(input())

print(binary_search(arr, target))
```


```
def binary_search(arr, target):
    arr.sort()

    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return True
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    return False


arr = list(map(int, input().split(',')))
target = int(input())
```

```
23  print(binary_search(arr, target))
```

Feedback

Input	Expected	Got
1,2,3,5,8 6	False	False
3,5,9,45,42 42	True	True
52,45,89,43,11 11	True	True

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 2

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer:(penalty regime: 0 %)

```
1  n = int(input())
2  arr = list(map(int, input().split()))
3
4  peaks = []
5
6  for i in range(n):
7      if (i == 0 or arr[i] >= arr[i - 1]) and (i == n - 1 or arr[i] >= arr[i + 1]):
8          peaks.append(arr[i])
9
10 print(*peaks)
```

```python
n = int(input())
arr = list(map(int, input().split()))

peaks = []

for i in range(n):
    if (i == 0 or arr[i] >= arr[i - 1]) and (i == n - 1 or arr[i] >= arr[i + 1]):
        peaks.append(arr[i])

print(*peaks)
```

Feedback

Input	Expected	Got
7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6
4 12 3 6 8	12 8	12 8

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 3

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer:(penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    balance = 0
    count = 0

    for ch in s:
        if ch == 'R':
            balance += 1
        else:
            balance -= 1

        if balance == 0:
            count += 1

    return count
```


```
def BalancedStrings(s):
    balance = 0
    count = 0

    for ch in s:
        if ch == 'R':
            balance += 1
        else:
            balance -= 1

        if balance == 0:
            count += 1

    return count
```


Feedback

Test	Expected	Got
`print(BalancedStrings('RLRRLLRLRL'))`	4	4
`print(BalancedStrings('RLLLLRRRLR'))`	3	3

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 4

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer:(penalty regime: 0 %)

Reset answer

```
def missingNumber(nums):
    n = len(nums)
    expected_sum = n * (n + 1) // 2
    actual_sum = sum(nums)

    return expected_sum - actual_sum
```


```
def missingNumber(nums):
    n = len(nums)
    expected_sum = n * (n + 1) // 2
    actual_sum = sum(nums)

    return expected_sum - actual_sum
```

Feedback

Test	Expected	Got
`print(missingNumber([3,0,1]))`	2	2
`print(missingNumber([0,1]))`	2	2
`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 5

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer:(penalty regime: 0 %)

```
S1 = input()
S2 = input()
N = int(input())

result = ""

for ch in S1:
    if ch in S2 and ch not in result:
        result += ch

    if len(result) == N:
        break

print(result)
```


```
S1 = input()
S2 = input()
N = int(input())

result = ""

for ch in S1:
    if ch in S2 and ch not in result:
        result += ch

    if len(result) == N:
        break

print(result)
```


Feedback

Input	Expected	Got
abcbde cdefghbb bcd 3	bcd	bcd

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 6

Incorrect
Mark 0.00 out of 1.00
Flag question

Question text

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer:(penalty regime: 0 %)

```
s = input()
words = s.split()

result = []

for word in words:
    if word.lower() != word[::-1].lower():
        result.append(word)

print(" ".join(result))
```


```
s = input()
words = s.split()

result = []

for word in words:
    if word.lower() != word[::-1].lower():
        result.append(word)

print(" ".join(result))
```


Feedback

Input	Expected	Got
Malayalam is my mother tongue	is my mother tongue	is my mother tongue

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect
Marks for this submission: 0.00/1.00.

Question 7

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience

enc

Sample Output 1

xpri

Answer:(penalty regime: 0 %)

```
s1 = input()
s2 = input()

result = ""

for ch in s1:
    if ch not in s2:
        result += ch

print(result)
```


```
s1 = input()
s2 = input()

result = ""

for ch in s1:
    if ch not in s2:
        result += ch

print(result)
```


Feedback

Input	Expected	Got
experience enc	xpri	xpri

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 8

Correct
Mark 1.00 out of 1.00

Flag question

Question text

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 No 4	

Answer:(penalty regime: 0 %)

```
n = int(input())
arr = list(map(int, input().split()))
k = int(input())

found = False

for i in range(n):
    for j in range(i + 1, n):
        if arr[i] + arr[j] == k:
            found = True
            break
    if found:
        break

print("Yes" if found else "No")
```


```
n = int(input())
arr = list(map(int, input().split()))
k = int(input())

found = False

for i in range(n):
    for j in range(i + 1, n):
        if arr[i] + arr[j] == k:
            found = True
            break
    if found:
        break
```

```
print("Yes" if found else "No")
```

Feedback

Input	Expected	Got
5 8 9 12 15 3 11	Yes	Yes
6 2 9 21 32 43 43 1 4	No	No
6 13 42 31 4 8 9 17	Yes	Yes

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 9

Correct
Mark 1.00 out of 1.00
Flag question

Question text

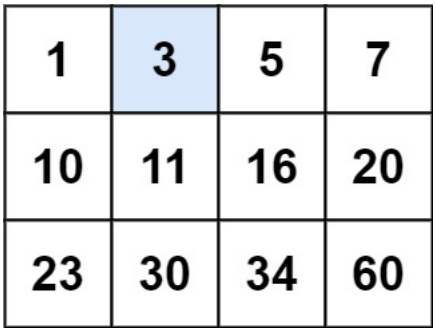
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:

1	3	5	7
10	11	16	20
23	30	34	60

Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False

For example:

Test	Result
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True

Answer:(penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix, target):
    m = len(matrix)
    n = len(matrix[0])

    low = 0
    high = m * n - 1

    while low <= high:
        mid = (low + high) // 2

        row = mid // n
        col = mid % n

        if matrix[row][col] == target:
            return True
        elif matrix[row][col] < target:
            low = mid + 1
        else:
            high = mid - 1

    return False
```


```
def searchMatrix(matrix, target):
    m = len(matrix)
    n = len(matrix[0])

    low = 0
    high = m * n - 1

    while low <= high:
        mid = (low + high) // 2

        row = mid // n
        col = mid % n

        if matrix[row][col] == target:
            return True
        elif matrix[row][col] < target:
            low = mid + 1
        else:
            high = mid - 1

    return False
```


Feedback

Test	Expected	Got
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False

Test	Expected	Got
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	`True`	`True`

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.

Question 10

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	`4`

Answer:(penalty regime: 0 %)

Reset answer

```
def search(nums, target):
    low = 0
    high = len(nums) - 1

    while low <= high:
        mid = (low + high) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    return -1
```


```
def search(nums, target):
    low = 0
    high = len(nums) - 1

    while low <= high:
        mid = (low + high) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
```

```
            low = mid + 1
        else:
            high = mid - 1

    return -1
```

Feedback

Test	Expected	Got
`print(search([-1,0,3,5,9,12],9))`	4	4
`print(search([-1,0,3,5,9,12],2))`	-1	-1

Passed all tests!

Correct
Marks for this submission: 1.00/1.00.
Finish review

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Status	Finished
Started	Thursday, 29 October 2026, 11:06 PM
Completed	Thursday, 29 October 2026, 11:09 PM
Duration	3 mins 2 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question 1

Correct
Mark 1.00 out of 1.00
Flag question

Question text

The average case occurs in the linear search algorithm

Question 1 Answer

a.

When the item is somewhere in the middle of the array

b.

When the item is the last element in the array

c.

Item is the last element in the array or item is not there at all

d.

When the item is not the array at all

Question 2

Correct
Mark 1.00 out of 1.00
Flag question

Question text

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

Question 2 Answer

a.

Selection

b.

Insertion

c.

Complexity

d.

Bubble

Question 3

Correct
Mark 1.00 out of 1.00
Flag question

Question text

What is the first step in binary search?

Question 3 Answer

a.
Divide the list into two equal parts

b.
Sort the list

c.
Compare the target element with the middle element in the list

d.
Compare the target element with the first element in the list

Question 4

Correct
Mark 1.00 out of 1.00
Flag question

Question text

In linear search, how is the element searched?

Question 4 Answer

a.
By comparing each element in the list sequentially

b.
By sorting the list first

c.
By using a hash function

d.
By dividing the list into halves

Question 5

Correct
Mark 1.00 out of 1.00
Flag question

Question text

What type of search would be most appropriate for finding an element in a list that is frequently updated?

Question 5 Answer

a.
Binary search

b.
Hash search

c.
Linear search

d.
Interpolation search

Question 6

Correct
Mark 1.00 out of 1.00
Flag question

Question text

_____ search takes a sorted/ordered list and divides it in the middle.

Question 6 Answer

a.

Hash

b.

Binary

c.

Both (1) & (3)

d.

Linear

Question 7

Correct
Mark 1.00 out of 1.00
Flag question

Question text

What is the key characteristic of binary search?

Question 7 Answer

a.
It always starts from the beginning of the list

b.
It works on unsorted lists

c.
It can be applied only if the list is sorted

d.
It compares elements sequentially

Question 8

Correct
Mark 1.00 out of 1.00
Flag question

Question text

In binary search, how is the middle element determined?

Question 8 Answer

a.
By dividing the list length by two

b.
By using a hash function

c.
By comparing each element sequentially

d.
By starting from the first element

Question 9

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Which method of searching involves sequentially comparing each element until a match is found?

Question 9 Answer

a.
Hashing

b.
Jump search

c.
Binary search

d.
Linear search

Question 10

Incorrect
Mark 0.00 out of 1.00
Flag question

Question text

What happens in a binary search if the list has an even number of elements?

Question 10 Answer

a.
The higher middle element is chosen as the middle element

b.
The search stops

c.
The middle element is chosen randomly

d.
The lower middle element is chosen as the middle element

Question 11

Correct
Mark 1.00 out of 1.00
Flag question

Question text

In binary search, what happens if the middle element does not match the target element?

Question 11 Answer

a.
The search continues in the left or right sublist

b.
The search stops

c.
The search continues from the beginning

d.
The list is sorted

Question 12

Correct

Mark 1.00 out of 1.00
Flag question

Question text

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

Question 12 Answer

a.
1

b.
n

c.
n/2

d.
log n

Question 13

Correct
Mark 1.00 out of 1.00
Flag question

Question text

During a binary search, what happens if the target element matches the middle element?

Question 13 Answer

a.
The search continues in the left sublist

b.
The search continues in the right sublist

c.
The list is sorted

d.
The search ends successfully

Question 14

Correct
Mark 1.00 out of 1.00
Flag question

Question text

Which of the following is a type of searching method?

Question 14 Answer

a.
Linear search

b.
Quick search

c.
Merge search

d.
Bubble search

Question 15

Correct
Mark 1.00 out of 1.00
Flag question

Question text

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

Question 15 Answer

a.
100

b.
10

c.
500

d.
1000

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